

## Written HW 4 — Full Solutions

Problem statements from Knight, *Physics for Scientists and Engineers* (5th ed., Global Edition), Chapter 25.

$\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ ,  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$ , and  $g = 9.8 \text{ m/s}^2$  throughout.

### 25.6 (2 pts)

What is the electric potential energy of the group of charges in Figure EX25.6? A  $-4.0 \text{ nC}$  charge and a  $+3.0 \text{ nC}$  charge sit  $3.0 \text{ cm}$  apart. A second  $+3.0 \text{ nC}$  charge sits  $4.0 \text{ cm}$  from the  $-4.0 \text{ nC}$  charge, at a right angle to the first pair, so the two  $+3.0 \text{ nC}$  charges are  $5.0 \text{ cm}$  apart.

**MODEL:** For a set of point charges, the total electric potential energy is the sum of the potential energies of every distinct pair.

**VISUALIZE:** Label the charges  $q_1 = -4.0 \text{ nC}$ ,  $q_2 = +3.0 \text{ nC}$ ,  $q_3 = +3.0 \text{ nC}$ , with  $r_{12} = 3.0 \text{ cm}$ ,  $r_{13} = 4.0 \text{ cm}$ , and  $r_{23} = 5.0 \text{ cm}$  (the hypotenuse of the 3-4-5 right triangle formed by the three charges). Three pairs, three terms.

**SOLVE:**

$$\begin{aligned} U &= \frac{1}{4\pi\epsilon_0} \left( \frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) \\ &= (8.99 \times 10^9) \left[ \frac{(-4.0 \times 10^{-9})(3.0 \times 10^{-9})}{0.030} + \frac{(-4.0 \times 10^{-9})(3.0 \times 10^{-9})}{0.040} + \frac{(3.0 \times 10^{-9})(3.0 \times 10^{-9})}{0.050} \right] \\ &= -3.60 \times 10^{-6} \text{ J} - 2.70 \times 10^{-6} \text{ J} + 1.62 \times 10^{-6} \text{ J} = \boxed{-4.7 \times 10^{-6} \text{ J}}. \end{aligned}$$

**REVIEW:** The two negative terms (each involving the  $-4.0 \text{ nC}$  charge paired with a  $+3.0 \text{ nC}$  charge) outweigh the one positive term (the two  $+3.0 \text{ nC}$  charges together), so the total comes out negative overall — consistent with the  $-4.0 \text{ nC}$  charge being closer to both  $+3.0 \text{ nC}$  charges than they are to each other.

## 25.8 (2 pts)

Three charged particles sit at the corners of an equilateral triangle with 4.0 cm edges. One particle has charge +4.0 nC, a second has charge +8.0 nC. What charge must the third particle have for the total electric potential energy of the three particles to be zero?

**MODEL:** Every side of an equilateral triangle has the same length, so all three pairwise separations equal  $d = 4.0$  cm. Set the sum of the three pairwise potential energies to zero and solve for the unknown charge  $q$ .

**VISUALIZE:** Call the known charges  $q_1 = +3.0$  nC and  $q_2 = +6.0$  nC and the unknown charge  $q$  — with both given charges positive, their pairwise energy  $U_{12}$  is already positive, so  $q$  must be negative to bring the total back down to zero.

**SOLVE:**

$$U = \frac{1}{4\pi\epsilon_0 d} [q_1 q_2 + q_1 q + q_2 q] = 0 \Rightarrow q_1 q_2 = -(q_1 + q_2) q$$
$$(3.0 \text{ nC})(6.0 \text{ nC}) = -(9.0 \text{ nC}) q \Rightarrow q = \frac{18 \text{ nC}^2}{-9.0 \text{ nC}} = \boxed{-2.0 \text{ nC}}.$$

**REVIEW:** Both given charges are positive, so the pair between them contributes a positive term no matter what. The only way to zero out the total is a negative third charge, and that's exactly what falls out — a good sign the setup is right before even checking the arithmetic.

### 25.19 (3 pts)

A proton travels along the  $x$ -axis at points where the electric potential is  $V(x) = (200 \text{ V/m}) x$ . As it passes  $x = 0 \text{ m}$ , the proton's speed is  $2.5 \times 10^5 \text{ m/s}$ . What speed does it have when it reaches  $x = 1.0 \text{ m}$ ?

**MODEL:** Energy is conserved. The proton's total mechanical energy, kinetic plus electric potential, is the same at both points.

**VISUALIZE:** Call  $x = 0$  point 1 and  $x = 1.0 \text{ m}$  point 2, so  $V_1 = 0 \text{ V}$  and  $V_2 = 200 \text{ V}$ . The proton is positive and moving into higher potential, so it should lose kinetic energy — we expect it to slow down.

**SOLVE:** Using  $U = qV$  and conservation of energy,

$$\begin{aligned}\frac{1}{2}mv_1^2 + qV_1 &= \frac{1}{2}mv_2^2 + qV_2 \Rightarrow v_2 = \sqrt{v_1^2 - \frac{2q}{m}(V_2 - V_1)} \\ &= \sqrt{(2.5 \times 10^5)^2 - \frac{2(1.60 \times 10^{-19})}{1.67 \times 10^{-27}}(200 \text{ V})} = \boxed{1.5 \times 10^5 \text{ m/s}}.\end{aligned}$$

**REVIEW:** The proton does slow down, as expected for a positive charge moving toward higher potential — the same rule from the height analogy: a positive charge climbing toward higher  $V$  trades kinetic energy for potential energy, just like a ball climbing a hill trades kinetic energy for gravitational PE.

## 25.26 (3 pts)

A proton is fired with speed  $2.00 \times 10^5$  m/s from the midpoint of a parallel-plate capacitor, straight toward the positive plate. The plates are held at 0 V and +500 V. (a) Does the proton reach the positive plate? (b) What speed does it have when it returns to the negative plate? *Also: do you have enough information to determine the  $x$ -position of the proton when it turns around? If so, what is that point? If not, what information is missing?*

**MODEL:** The potential inside a parallel-plate capacitor rises linearly from the negative plate to the positive plate, so the midpoint sits exactly halfway between the two plate voltages:  $V_{\text{mid}} = 250$  V. Energy is conserved throughout.

**VISUALIZE:** The proton starts at the midpoint moving toward the positive (higher- $V$ ) plate, so it decelerates. Whether it reaches the plate depends on whether its initial kinetic energy is enough to supply the potential-energy increase over the remaining  $500 \text{ V} - 250 \text{ V} = 250 \text{ V}$  it would still have to climb.

**SOLVE:** (a) The kinetic energy available is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(1.67 \times 10^{-27} \text{ kg})(2.00 \times 10^5 \text{ m/s})^2 = 3.34 \times 10^{-17} \text{ J}.$$

Reaching the positive plate requires climbing the remaining 250 V from the midpoint, which costs

$$\Delta U = e \Delta V = (1.60 \times 10^{-19} \text{ C})(250 \text{ V}) = 4.00 \times 10^{-17} \text{ J}.$$

Since  $K < \Delta U$ , the proton does not reach the positive plate — it runs out of kinetic energy first and turns around.

(b) The proton returns through the midpoint with its original speed (energy conservation is symmetric) and continues on to the negative plate, which is at *lower* potential than the midpoint, so it speeds up over that stretch:

$$v_f = \sqrt{v^2 + \frac{2e}{m}(250 \text{ V} - 0 \text{ V})} = \boxed{2.96 \times 10^5 \text{ m/s}}.$$

**Turning point:** Yes, this is determined. At the instant it turns around, all of its initial kinetic energy has converted to potential energy, so the potential there,  $V_{\text{turn}}$ , satisfies  $e(V_{\text{turn}} - 250 \text{ V}) = K = 3.34 \times 10^{-17} \text{ J}$ , giving  $V_{\text{turn}} = 250 \text{ V} + 209 \text{ V} = 459 \text{ V}$ . Since  $V$  rises linearly with position, that's  $459/500 = 91.8\%$  of the way from the negative plate to the positive plate — *as a fraction of the full plate separation*. Converting that fraction into an actual distance (in mm, say) would need the plate spacing itself, which isn't needed for anything else in this problem and may or may not be given in the figure; if it is, multiply the plate spacing by 0.918 to get the position.

**REVIEW:** Parts (a) and (b) both hinge on the same idea from lecture:  $V$  doesn't have to change by equal amounts to change the outcome, but here it changes *steadily*, so distance and potential are directly proportional — which is exactly what makes the turning-point fraction (91.8% of the gap) translate directly into a position, unlike a capacitor with a bumpier, non-uniform  $V(x)$ .

### 25.30 (2 pts)

What is the electric potential at the point indicated with the dot in Figure EX25.30? Three charges  $+1.0\text{ nC}$ ,  $+2.0\text{ nC}$ , and  $-2.0\text{ nC}$  sit at the corners of a symmetric arrangement, each a horizontal distance of  $1.5\text{ cm}$  from the dot's vertical line, at  $30^\circ$  from the line to the dot.

**MODEL:** The net potential at a point is the scalar sum of the potentials from each charge — no components, no angles needed once each distance to the dot is known.

**VISUALIZE:** By the  $30^\circ$  geometry, all three charges are the same distance  $r$  from the dot:  $r = (1.5\text{ cm})/\cos 30^\circ = 1.732\text{ cm}$ .

**SOLVE:**

$$V = \frac{1}{4\pi\epsilon_0} \left( \frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} \right) = (8.99 \times 10^9) \left[ \frac{1.0 \times 10^{-9}}{0.01732} + \frac{2.0 \times 10^{-9}}{0.01732} + \frac{-2.0 \times 10^{-9}}{0.01732} \right]$$
$$= \boxed{+520\text{ V}}.$$

**REVIEW:** Since  $r$  is the same for all three charges, the whole sum collapses to  $\frac{1}{4\pi\epsilon_0 r}(q_1 + q_2 + q_3)$  — the net potential of three equidistant charges only cares about their *total* charge,  $+1.0\text{ nC}$  in this case. That shortcut only works because all three distances happen to be equal; it would not apply if the charges sat at different distances from the dot.

### 25.38 (5 pts)

A  $-10.0$  nC point charge and a  $+20.0$  nC point charge are  $15.0$  cm apart on the  $x$ -axis, with the  $+20.0$  nC charge at the origin. (a) What is the electric potential at the point on the  $x$ -axis where the electric field is zero? (b) At what point on the  $x$ -axis is the electric potential zero, and what is the magnitude of the electric field at that point?

**MODEL:** The point where  $\vec{E} = 0$  is found from the field magnitudes of the two charges; the point where  $V = 0$  is found separately from the (scalar) potentials. These are two different points in general — nothing requires them to coincide.

**VISUALIZE:** Call  $Q_1 = +20.0$  nC (at  $x = 0$ ) and  $Q_2 = -10.0$  nC (at  $x = 15.0$  cm). Since  $Q_1$  is bigger, the field-canceling point must be on the far side of the smaller charge  $Q_2$ , at  $x > 15.0$  cm, where  $Q_1$ 's field (pointing away from  $Q_1$ ) and  $Q_2$ 's field (pointing toward  $Q_2$ ) point in opposite directions and can balance.

**SOLVE:** (a) Setting the two field magnitudes equal at position  $x$ :

$$\frac{1}{4\pi\epsilon_0} \frac{20.0 \text{ nC}}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{10.0 \text{ nC}}{(x - 15.0 \text{ cm})^2} \Rightarrow x^2 = 2(x - 15.0 \text{ cm})^2$$

$$\Rightarrow x^2 - (60.0 \text{ cm})x + 450 \text{ cm}^2 = 0 \Rightarrow x = 51.2 \text{ cm} \text{ (the other root, } 8.8 \text{ cm, falls between the charges and doesn't fit)}$$

The potential there is

$$V = \frac{1}{4\pi\epsilon_0} \left( \frac{20.0 \times 10^{-9}}{0.512} + \frac{-10.0 \times 10^{-9}}{0.512 - 0.150} \right) = \boxed{+103 \text{ V}}.$$

(b) Setting the potentials to cancel instead, at a point  $x$  between the charges:

$$\frac{20.0 \text{ nC}}{x} + \frac{-10.0 \text{ nC}}{15.0 \text{ cm} - x} = 0 \Rightarrow 2(15.0 \text{ cm} - x) = x \Rightarrow \boxed{x = 10.0 \text{ cm}}.$$

The field there is the vector sum of both contributions (both pointing in  $+\hat{x}$  at this point, since it's to the right of  $Q_1$  and to the left of  $Q_2$ ):

$$E = \frac{1}{4\pi\epsilon_0} \frac{20.0 \times 10^{-9}}{(0.100)^2} + \frac{1}{4\pi\epsilon_0} \frac{10.0 \times 10^{-9}}{(0.050)^2} = \boxed{5.40 \times 10^4 \text{ N/C}}.$$

**REVIEW:** The two answers land on genuinely different points ( $51.2$  cm vs.  $10.0$  cm) — a direct illustration that  $E = 0$  and  $V = 0$  are unrelated conditions. At the field-zero point the potential is a nonzero  $+103$  V; at the potential-zero point the field is a very much nonzero  $5.4 \times 10^4$  N/C. Knowing one tells you nothing about the other.